Mathematical Background and Reading Guide

for Constrained 2DOF PIDF Tuning

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These notes accompany *Constrained Setpoint-Zero Shaping in Two-Degree-of-Freedom PIDF Control: Modal Suppression, Monotone Tracking, and MIACE with Pure Delay* [1]. They develop the additional mathematical tools needed after the MIACE proofs companion [2]: affine coefficient geometry, modal bounds, convex optimisation, compactness of the monotone class, and response moments. The final reading map covers all 18 numbered results of the paper.

1 The question behind the calculations

A controller may reject disturbances well and still react badly to a change of reference. Changing its feedback gains changes both behaviours. Setpoint weighting leaves those gains in place and changes the signal which drives the existing loop. There are only two weights, b and c, but their effect is a whole function of time. The main question is how to choose two numbers while controlling that entire function.

For a fixed feedback loop, the answer begins with an affine dependence. At each time, the output and command are affine functions of the weights. At a fixed pole, the residue is affine as well. A sign constraint at every time becomes an intersection of half-planes in the weight plane; a bound on a complex residue becomes a cone constraint. The different parts of the paper follow from these two elementary observations, together with the stability needed to integrate the resulting signals.

The background companion [2], denoted below by C13, supplies the integration and convolution arguments. Its Chapter 2 explains absolute areas, Tonelli's theorem and absolute continuity. Chapters 3–5 treat positive convolution kernels, Laplace transforms and retained internal states. Section 6.1 derives the signed-area balance and the MIACE lower bound. The present guide takes those results as its starting point. Calculus, finite-dimensional linear algebra and elementary complex numbers suffice for the remaining calculations developed here.

"Paper Theorem 14" always means a statement in the accompanying manuscript. Numbered equations and results without that prefix belong to this guide. The reading map in Section 8 refers to the 18-result MCSS version prepared on 23 October 2026. Its notation is retained: T_i, T_d, T_f are positive controller time constants; $K_p > 0$ is the proportional gain; $K_i = K_p/T_i$ and $K_d = K_p T_d$. All step responses used below start from zero states and zero prehistory. Stability of an implemented system includes its retained states and, when there is a delay, its admissible initial histories.

2 What remains fixed, and what the weights can assign

For the derivative-branch filter, the reference and feedback controllers are

$$C_r = K_p\left(b + \frac{1}{T_i s} + \frac{cT_d s}{1+T_f s}\right), \qquad C_y = K_p\left(1 + \frac{1}{T_i s} + \frac{T_d s}{1+T_f s}\right). \tag{2.1}$$

The same integral state implements the error integral in both expressions. With a filter state x_F, the actual equations are

$$\begin{aligned} \dot{x}_I &= r - y, \qquad T_f \dot{x}_F = cr - y - x_F, \\ u &= K_p \left[br - y + x_I/T_i + (T_d/T_f)(cr - y - x_F) \right]. \end{aligned} \tag{2.2}$$

Setting $r = 0$ removes both weights from these equations. Thus changing b, c changes the forcing of the state system, while its homogeneous dynamics stay fixed. This observation is also valid when the plant contains a transport delay: the same delay and the same history operator remain in the homogeneous equations.

For a plant P, put $S = (1 + PC_y)^{-1}$. A disturbance d at the plant input and noise n added to the measurement give

$$\begin{pmatrix} y \\ u \end{pmatrix} = \begin{pmatrix} PC_rS \\ C_rS \end{pmatrix} r + \begin{pmatrix} PS & -PC_yS \\ -C_yPS & -C_yS \end{pmatrix} \begin{pmatrix} d \\ n \end{pmatrix}. \tag{2.3}$$

The second matrix contains no reference weight. Its invariance is exact; no comparison of sampled frequency plots is needed. The reference transfer $G = PC_rS$ does change. In particular, the reference error transfer is $1 - G$, whereas the usual sensitivity is S. They coincide at $b = c = 1$, but need not coincide at other weights.

An external cancellation has a similarly precise meaning. If a stable state is no longer driven from the reference, it may still be driven by a disturbance or excited by its initial condition. C13 Section 5.3 gives an elementary example of the distinction between a transfer function and its retained realisation. Paper Theorem 6 uses polynomial cancellation only after internal stability has been assumed.

2.1 A rectangle of weights becomes a coefficient region

On the denominator $T_i s(1 + T_f s)$ the reference numerator is

$$N_D(s) = 1 + (T_f + bT_i)s + T_i(bT_f + cT_d)s^2. \tag{2.4}$$

To assign $Q(s) = 1 + \sigma s + \pi s^2$, compare the two coefficients:

$$\begin{pmatrix} \sigma \\ \pi \end{pmatrix} = \begin{pmatrix} T_f \\ 0 \end{pmatrix} + \begin{pmatrix} T_i & 0 \\ T_iT_f & T_iT_d \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix}. \tag{2.5}$$

The determinant is $T_i^2 T_d > 0$, so every target coefficient pair has exactly one real preimage. Solving the first row and then the second gives

$$b = \frac{\sigma - T_f}{T_i}, \qquad c = \frac{\pi - T_f(\sigma - T_f)}{T_iT_d}. \tag{2.6}$$

Admissibility is checked by putting these values into the allowed intervals. An affine map takes each segment to a segment, so the image of a rectangular weight box is a parallelogram. Filtering the entire controller output instead gives $N_O = 1 + bT_i s + cT_iT_d s^2$ and a rectangular coefficient region. This second placement also changes C_y; it defines a different fixed feedback loop.

The command-jump formula is obtained at the other end of the frequency axis. A strictly proper plant has no instantaneous output response. For a unit reference step, the right-hand command value is therefore $C_r(\infty)$. In the two placements it is respectively $K_p(b + cT_d/T_f)$ and K_pcT_d/T_f. Substitution of the appropriate inverse coefficient map gives, in both cases,

$$u(0^+) = \frac{K_p\pi}{T_iT_f}. \tag{2.7}$$

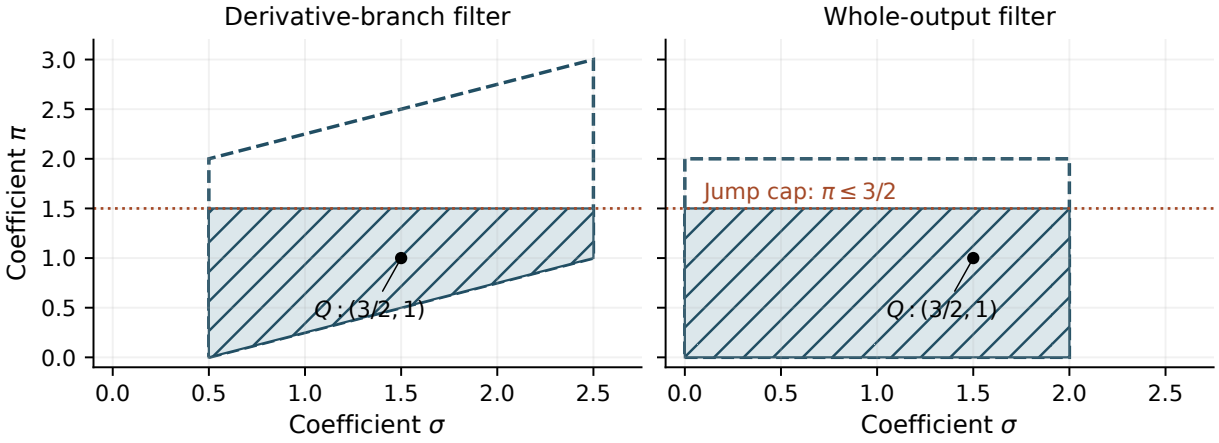


Figure S1: The worked coefficient maps. Dashed boundaries show the full image of the weight box; shading additionally imposes $\pi \leq 3/2$. The marked target is feasible in both placements, with different weights.

For a step of amplitude Δr, multiply this expression by Δr. A jump cap cuts the coefficient region by a horizontal line. A stable complex pair with natural frequency ω_z and damping ratio ζ_z has $\pi = \omega_z^{-2}$ and $\sigma = 2\zeta_z/\omega_z$. Combining the jump cap with the D-branch restriction $\sigma \geq T_f$ yields Paper Corollary 5. The bound concerns the initial jump; a later command peak requires another argument.

Example 2.1 (Reading the inverse map)**.** Take $T_i = 2$, $T_d = 1$, $T_f = 1/2$, $K_p = 1$ and $0 \leq b, c \leq 1$. For $Q(s) = 1 + \frac{3}{2}s + s^2$, the D-branch weights are $(b, c) = (1/2, 1/4)$ and the output-filter weights are $(3/4, 1/2)$. Both produce $u(0^+) = 1$. Figure S1 shows the two feasible regions and the additional cap $u(0^+) \leq 3/2$. This coefficient calculation asserts assignability and jump size; stability must still be checked for each feedback realisation.

2.2 How a parameter-box stability proof works

For a quartic $a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0$ with all $a_j > 0$, define

$$H_2 = a_3 a_2 - a_4 a_1, \qquad H_3 = a_1 H_2 - a_3^2 a_0. \tag{2.8}$$

When H_2, H_3 are nonzero, the first column of the Routh array is

$$a_4, \quad a_3, \quad H_2/a_3, \quad H_3/H_2, \quad a_0. \tag{2.9}$$

The Routh criterion makes positivity of this column equivalent to all roots lying in the open left half-plane. The criterion itself is a standard algebraic stability result; the family-specific work is proving its inequalities throughout the parameter set.

Paper Proposition 12 considers $P(s) = K/(T^2 s^2 + 2\zeta T s + 1)$, with $K, T > 0$ and $\zeta \geq 0$, and collects powers of the nonnegative quantity $v = 2\zeta T$. With $f = T_f$, $i = T_i$, $d = T_d$, $\kappa = KK_p$, $C = d - f^2/i$ and $\gamma = 1 + f/i$, the constant term of H_3 in v is

$$\kappa T^2 \{C(1 + \kappa\gamma) - T^2/i\}. \tag{2.10}$$

Under $f \leq i$ and $C > 0$, all other coefficients in that expansion are nonnegative. Thus $T_{\max}^2 < iC(1 + \kappa_{\min}\gamma)$ makes this term strictly positive for every admitted T, κ, and the corresponding expansion gives $H_2 > 0$. In the paper's example the comparison is exactly $144 < 222.59904$. The decimals are rational numbers, so the inequality can be checked without computing roots. This establishes stability across the family. Simplicity of a selected pole and bounds on its residue require their own hypotheses.

3 From a pole residual to a whole response

Suppose a rational reference transfer is $T(s) = H(s)N(s)/D(s)$ and $p \neq 0$ is a simple stable pole. Near p, write $D(s) = (s-p)D'(p) + O((s-p)^2)$. Multiplication by $s-p$ and passage to the limit give

$$r_p = \operatorname{Res}_{s=p} T(s) = \frac{H(p)N(p)}{D'(p)}. \tag{3.1}$$

The factor $N(p)$ measures the failure of cancellation, but $D'(p)$ also matters. Two almost coincident poles can make this denominator small. That is the reason for the separation assumptions in Paper Corollary 8.

For a nominal pair $\widehat{p}, \overline{\widehat{p}}$, normalise its factor by its value at zero:

$$q(s) = \frac{(s-\widehat{p})(s-\overline{\widehat{p}})}{|\widehat{p}|^2}. \tag{3.2}$$

At a displaced pole $p = \widehat{p} + e$,

$$q(p) = \frac{e(2i\operatorname{Im}\widehat{p} + e)}{|\widehat{p}|^2}, \qquad |q(p)| \leq \frac{\varepsilon(2|\operatorname{Im}\widehat{p}| + \varepsilon)}{|\widehat{p}|^2} \quad (|e| \leq \varepsilon). \tag{3.3}$$

The upper bound is attained when the two complex terms align and $|e| = \varepsilon$. A particular uncertain plant may trace only a curve inside the disk, so it need not attain that disk bound.

For a unit step, the transformed output is $T(s)/s$. Its residue at p is $a_p = r_p/p$. A conjugate pair contributes

$$a_p e^{pt} + \overline{a_p} e^{\bar{p}t} = 2e^{\operatorname{Re} p\, t} \operatorname{Re}(a_p e^{i \operatorname{Im} p\, t}), \tag{3.4}$$

whose modulus is at most $2|r_p/p|e^{\operatorname{Re} p\, t}$. Division by p is essential. A mode with $|r_p| = 0.1$ and $|p| = 0.2$ has step-pair envelope coefficient 1; one with $|r_p| = 0.2$ and $|p| = 2$ has coefficient 0.2. Ranking the residues alone gives the opposite order. The decay rates must then be considered as well.

3.1 Closing the time axis

For a stable strictly proper rational transfer with unit DC gain and simple poles $p_1, \ldots, p_n$, the full step response is

$$y(t) = 1 + \sum_{j=1}^{n} a_j e^{p_j t}. \tag{3.5}$$

Both members of each complex pair are included. The triangle inequality gives

$$|1 - y(t)| \leq \sum_j |a_j| e^{\operatorname{Re} p_j t}, \tag{3.6}$$

$$\int_H^\infty |1 - y(t)|\, \mathrm{d}t \leq \sum_j \frac{|a_j|}{-\operatorname{Re} p_j} e^{\operatorname{Re} p_j H}. \tag{3.7}$$

The first bound decreases with time. If it is below a tracking tolerance at H, it remains so thereafter. The second bounds the unobserved error area after a finite simulation. Neither controls quadrature error on the observed interval.

For the command $u(t) = u_\infty + \sum_j d_j e^{p_j t}$, the same argument bounds its tail by $|u_\infty| + \sum_j |d_j| e^{\operatorname{Re} p_j H}$. At a repeated pole, terms such as te^{pt} must be retained. For a genuine delayed feedback system, a finite rational modal expansion is generally unavailable. The delayed example in Section 7 uses a convolution argument instead; the finite sums here belong to the rational part of the paper.

On a finite interval $[t_k, t_{k+1}]$ of length h_k, suppose $|\dot{u}| \le L_k$. Every point lies within $h_k/2$ of an endpoint, and absolute continuity gives

$$|u(t)| \le |u(t_e)| + L_k|t - t_e|. \tag{3.8}$$

Consequently, imposing $|u(t_k)|, |u(t_{k+1})| \le \bar{u} - L_k h_k/2$ proves $|u(t)| \le \bar{u}$ throughout the interval. For instance, $\bar{u} = 3$, $h_k = 0.1$ and $L_k = 2$ require endpoint bounds of 2.9. If a command bound is active at $t = 0^+$, this symmetric margin can fail even when the command immediately decreases. A derivative-sign argument can then give a sharper bound, as in Paper Section 7.1.

For a weight polygon with vertices $\mathcal{V}$, put $M_j = \max_{w\in\mathcal{V}} |d_j(w)|$. Affinity of d_j and the triangle inequality imply $|d_j(w)| \le M_j$ everywhere in the polygon. One may therefore use the weight-independent bound

$$L_k = \sum_j |p_j| M_j e^{\mathrm{Re}\, p_j t_k}. \tag{3.9}$$

The tightened endpoint constraints remain affine in b, c. Together with the tail bound, they provide a sufficient whole-time bound for each declared scenario. Saturation and anti-windup dynamics are outside this linear calculation.

4 Why the design problem is convex

Let $w = (b, c)$ and let $z_j(w) = q_{j0} + q_{jb}b + q_{jc}c$ be a complex modal amplitude. For $0 \le \lambda \le 1$,

$$|z_j(\lambda w + (1-\lambda)\widetilde{w})| \le \lambda|z_j(w)| + (1-\lambda)|z_j(\widetilde{w})|. \tag{4.1}$$

This is just the triangle inequality applied after the affine identity. Taking the largest amplitude preserves the inequality, so $F(w) = \max_j |z_j(w)|$ is convex. On a nonempty compact admitted polygon X, continuity then guarantees an attained minimum.

Write the real and imaginary parts as $A_j w + d_j \in \mathbb{R}^2$. Introducing one scalar t gives the equivalent problem

$$\min_{w\in X, t} t \quad \text{subject to} \quad \|A_j w + d_j\|_2 \le t \quad \text{for every } j. \tag{4.2}$$

Each constraint says that $(t, A_j w + d_j)$ belongs to the second-order cone $\{(t, x) : \|x\|_2 \le t\}$. Thus the cone formulation expresses the modulus exactly. It does not approximate the complex plane by separate bounds on real and imaginary parts. The same construction works for impulse residues and step amplitudes; their coefficients differ by the factor $1/p_j$.

4.1 A lower bound from a supporting line

A vector g is a subgradient of a convex function f at w_* if $f(w) \ge f(w_*) + g^T(w - w_*)$ for every w. For $f(w) = |z(w)|$ and $z(w_*) \ne 0$, differentiation gives

$$g = \frac{1}{|z(w_*)|} \begin{pmatrix} \mathrm{Re}(\overline{z(w_*)} q_b) \\ \mathrm{Re}(\overline{z(w_*)} q_c) \end{pmatrix}. \tag{4.3}$$

At a zero, $g = 0$ is valid because $f \ge 0$. Choose nonnegative multipliers λ_j summing to one. Averaging the supporting inequalities gives

$$F(w) \ge \sum_j \lambda_j f_j(w_*) + \left(\sum_j \lambda_j g_j\right)^T (w - w_*). \tag{4.4}$$

The minimum of the right-hand affine function over a polygon occurs at a vertex. Indeed, any point is a convex combination of vertices, and its affine value is the same combination of their values. Therefore

$$\begin{aligned} L &= \sum_j \lambda_j f_j(w_*) + \min_{v\in\mathcal{V}} \left(\sum_j \lambda_j g_j\right)^T (v - w_*), \\ L &\le F^* \le F(w_*) \qquad (w_* \in X). \end{aligned} \tag{4.5}$$

Any admissible multipliers give a mathematical lower bound. Choosing them well makes it sharper.

Example 4.1 (A minimax point without a solver)**.** On $0 \le x \le 1$, minimise $F(x) = \max\{x, 2(1-x)\}$. The two branches meet at $x_* = 2/3$. Their slopes are 1 and -2; weights $2/3$ and $1/3$ make the averaged slope zero. The averaged supporting line is the constant $2/3$, so $F(x) \ge 2/3$ everywhere. The crossing point attains that bound. The paper uses precisely this supporting-plane argument in two dimensions, with moduli in place of straight branches.

Floating-point evaluation of such a bound remains a numerical check. A gap near machine precision does not enclose the exact answer unless errors in poles, amplitudes and gradients have also been bounded. The general convex framework can be studied further in Boyd and Vandenberghe [4].

4.2 What a parameter grid can establish

Fix the parameter coordinates and their scaling. Suppose the selected complex amplitude $z_\theta(w)$ has a uniform derivative bound $\|D_\theta z_\theta(w)\|_{\mathrm{op}} \le L_\Theta$ on a compact convex parameter set, for every $w \in X$. If the sample points have fill distance h_Θ, integration along the segment to a nearest sample gives

$$\max_j |z_{\theta_j}(w)| \le \sup_{\theta\in\Theta} |z_\theta(w)| \le \max_j |z_{\theta_j}(w)| + L_\Theta h_\Theta. \tag{4.6}$$

The same additive bound separates the grid and continuum optimal values, provided both use the same admitted set X. A selected simple pole satisfies

$$\partial_{\theta_k} p_\theta = -\frac{\partial_{\theta_k} D_\theta(p_\theta)}{\partial_s D_\theta(p_\theta)}. \tag{4.7}$$

This identifies what must be bounded to obtain L_Θ: the pole branch, its separation, the denominator derivative and the remaining amplitude factors throughout the parameter set. Values checked only at sample points do not establish that derivative bound. Paper Proposition 12 supplies continuum stability; it does not supply these additional continuum amplitude estimates.

5 Why the monotone feasible set is compact

A transient can have small oscillatory residues and still turn backwards. To exclude that motion, Paper Theorem 14 imposes a sign on the whole impulse response. The resulting constraint is infinite in time, but its unknowns remain the two real numbers b, c. This combination is what makes an existence proof possible even when the admitted weight set is unbounded.

Retain the derivative-branch filter and the shared-state controller. Let

$$P_L(s) = e^{-Ls} P_0(s), \qquad P_0(0) = k > 0, \qquad K_i = K_p/T_i, \quad K_d = K_p T_d,$$

where P_0 is strictly proper and real rational, with no pole at zero. The fixed implemented feedback loop is internally exponentially stable, including its admissible initial histories. The reference experiment starts with zero states and zero prehistory. These assumptions supply ordinary integrable

output kernels and their moments. They allow unstable poles in the open-loop plant; stability is required of the implemented closed loop.

With $S_L = (1 + P_L C_y)^{-1}$, separate the three reference contributions:

$$G_0 = P_L S_L \frac{K_i}{s}, \qquad G_b = P_L S_L K_p, \qquad G_c = P_L S_L \frac{K_d s}{1 + T_f s}. \tag{5.1}$$

Their impulse responses belong to $L^1(0, \infty)$, and

$$g_{b,c} = g_0 + b g_b + c g_c, \qquad y_{b,c}(t) = \int_0^t g_{b,c}(r)\, \mathrm{d}r. \tag{5.2}$$

The subscripts on g_b, g_c name basis functions. Integral action gives $G_{b,c}(0) = 1$ for every finite pair of weights, hence $\int g_{b,c} = 1$. The basis variations themselves have zero integral.

5.1 A sign condition in a function space

The step response is locally absolutely continuous. Therefore it is nondecreasing exactly when $g_{b,c} \geq 0$ almost everywhere. In one direction, integrate the nonnegative derivative between any two times. In the other, the derivative of a nondecreasing function is nonnegative wherever it exists, and absolute continuity identifies that derivative with the kernel almost everywhere. Together with unit mass, this gives $0 \leq y_{b,c}(t) \leq 1$ for every t, and $y_{b,c}(t) \to 1$.

An almost-everywhere kernel statement thus gives an every-time statement about the trajectory. A kernel may jump at a delay boundary; assigning a different value at that isolated point changes neither its integral nor the output. Requiring a particular point value would add a condition unrelated to the physical response.

Let $X \subseteq \mathbb{R}^2$ be closed and convex, and define

$$\mathcal{F} = \{(b, c) \in X : g_0 + b g_b + c g_c \geq 0 \text{ almost everywhere}\}.$$

To see that this set is closed, suppose nonnegative functions f_n converge to f in L^1. At a point where $f < 0$, nonnegativity of f_n gives $-f \leq |f - f_n|$. Consequently

$$\int_0^\infty \max(-f, 0)\, \mathrm{d}t \leq \|f - f_n\|_1 \longrightarrow 0.$$

The negative part of f vanishes almost everywhere. This proves that the positive cone of L^1 is closed. It is convex because a convex combination of nonnegative functions remains nonnegative. The affine map in (5.2) is continuous, since

$$\|g_{b,c} - g_{\widetilde{b},\widetilde{c}}\|_1 \leq |b - \widetilde{b}|\, \|g_b\|_1 + |c - \widetilde{c}|\, \|g_c\|_1.$$

Taking the inverse image of the cone and intersecting with X proves closedness and convexity of $\mathcal{F}$.

5.2 Why the weights cannot escape to infinity

Closedness alone does not give a minimiser on an unbounded set. Here positivity supplies the missing bound. Every feasible kernel has unit mass, so

$$\|b g_b + c g_c\|_1 = \|g_{b,c} - g_0\|_1 \leq 1 + \|g_0\|_1. \tag{5.3}$$

This bounds a function. To obtain a bound on its two coefficients, we must first exclude cancellation between dependent basis functions.

Suppose $\beta g_b + \gamma g_c = 0$ in L^1. Its Laplace transform is zero. On an open set where the common nonzero factor $P_L S_L$ does not vanish, (5.1) gives

$$\beta K_p + \gamma \frac{K_d s}{1 + T_f s} = 0.$$

Multiplication by $1 + T_f s$ gives the polynomial identity

$$\beta K_p + (\beta K_p T_f + \gamma K_d) s = 0.$$

Since $K_p, K_d > 0$, its two coefficients force $\beta = \gamma = 0$. Thus g_b, g_c are linearly independent.

The continuous function $(\beta, \gamma) \mapsto \|\beta g_b + \gamma g_c\|_1$ now has a positive minimum η on the Euclidean unit circle. Compactness gives a minimum; independence prevents that minimum from being zero. Rescaling yields

$$\|b g_b + c g_c\|_1 \geq \eta \sqrt{b^2 + c^2}.$$

Combining this inequality with (5.3) bounds both weights. Hence $\mathcal{F}$ is closed and bounded in $\mathbb{R}^2$, and therefore compact. If it is nonempty, every continuous objective attains a minimum on it. In particular, the largest feasible b exists. No compactness theorem for an infinite-dimensional unit ball is involved: the kernels vary in a fixed two-dimensional affine family.

For PI control, $K_d = 0$ makes $g_c = 0$. The redundant coordinate c must then be removed before making this compactness assertion. The remaining one-dimensional argument is unchanged.

5.3 The reference norm and the feedback norms

For a feasible kernel, the triangle inequality gives, at every real frequency,

$$|G_{b,c}(\mathrm{i}\omega)| = \left| \int_0^\infty e^{-\mathrm{i}\omega t} g_{b,c}(t) \, \mathrm{d}t \right| \leq \int_0^\infty g_{b,c}(t) \, \mathrm{d}t = 1.$$

At zero frequency equality holds. Thus $\|G_{b,c}\|_\infty = 1$. The same integral estimate holds in the open right half-plane, and its limit at zero gives the same supremum. The exact transport delay is already contained in $g_{b,c}$.

Paper Proposition 13 preserves the feedback norms for another reason: setting the reference to zero removes b, c from the shared-state equations. The disturbance and measurement-noise operators are identical for every pair of weights. The reference operator changes, but positivity and unit gain fix its norm throughout $\mathcal{F}$. Its command counterpart $U_{b,c}$ has no such general norm identity.

Finally, monotone output does not establish the extra command sign used in the MIACE projection. For the normalised plant $y' + y = u$, choose $y = 1 - e^{-at}$ with $0 < a < 1$. Then

$$u = 1 + (a - 1)e^{-at}, \qquad v = u - 1 = (a - 1)e^{-at} < 0,$$

although y increases from zero to one. The set $\mathcal{F}_+$ in Paper Theorem 17 therefore imposes a substantive additional constraint. Its command signs require their own proof.

6 What the response moments measure

Suppose that an impulse response g is nonnegative, has unit mass, and has the exponential decay supplied by the fixed stable feedback system. The step response and its remaining error are

$$y(t) = \int_0^t g(r) \, \mathrm{d}r, \qquad e(t) = 1 - y(t) = \int_t^\infty g(r) \, \mathrm{d}r.$$

At time t, the error is the mass which has yet to arrive. Integrating that error counts each portion of mass for as long as it is waiting. Tonelli's theorem therefore gives

$$\int_0^\infty e(t)\ \mathrm{d}t = \int_0^\infty \int_0^r 1\ \mathrm{d}t\, g(r)\ \mathrm{d}r = \int_0^\infty r g(r)\ \mathrm{d}r. \tag{6.1}$$

The same calculation with the weight t gives

$$\int_0^\infty te(t)\ \mathrm{d}t = \frac{1}{2}\int_0^\infty r^2 g(r)\ \mathrm{d}r. \tag{6.2}$$

Companion [2], Sections 2.4 and 4.2, supplies the integration and transform rules. Here they explain the particular optimisation in Paper Theorem 14 and Proposition 15.

Write $K_i = K_p/T_i$, $K_d = K_p T_d$, and

$$\Gamma = L - \frac{P_0'(0)}{k}, \qquad k = P_0(0) > 0.$$

Near zero, $P_L^{-1} = k^{-1} + \Gamma s/k + O(s^2)$. Since

$$1 - G_{b,c} = \frac{P_L^{-1} + C_y - C_r}{P_L^{-1} + C_y},$$

multiplication of numerator and denominator by s yields

$$1 - G_{b,c}(s) = \frac{n_0 s + n_1 s^2 + O(s^3)}{K_i + d_0 s + O(s^2)}, \quad \begin{cases} n_0 = k^{-1} + (1-b)K_p, \\ n_1 = \Gamma/k + (1-c)K_d, \\ d_0 = k^{-1} + K_p. \end{cases} \tag{6.3}$$

Ordinary division of these series gives

$$G_{b,c}(s) = 1 - ms + q_2 s^2 + O(s^3), \qquad m = \frac{n_0}{K_i}, \quad q_2 = \frac{d_0 m - n_1}{K_i}. \tag{6.4}$$

On the monotone feasible set $\mathcal{F}$, these coefficients are the two integrals above. In particular,

$$\mathrm{IAE} = m(b) = T_i\left(1 - b + \frac{1}{kK_p}\right), \qquad q_2 = \frac{(K_p + k^{-1})m(b) - (1-c)K_d - \Gamma/k}{K_i}. \tag{6.5}$$

Increasing b decreases the first moment. Once its largest feasible value b_* has been selected, increasing c increases the second moment, because $\partial q_2/\partial c = K_d/K_i > 0$. Thus the successive choice is largest feasible b, then smallest feasible c on that vertical slice. The first decision minimises IAE; the second resolves its remaining freedom by penalising late error more heavily.

The probability-density interpretation also gives a useful necessary condition. The variance is nonnegative:

$$2q_2 - m^2 = \int_0^\infty (t-m)^2 g(t)\ \mathrm{d}t \geq 0.$$

Substitution into (6.5) gives the lower bound on c in Paper Proposition 15:

$$c \geq 1 + \frac{K_i m(b)^2/2 - (K_p + k^{-1})m(b) + \Gamma/k}{K_d}. \tag{6.6}$$

A moment inequality alone does not establish positivity of the impulse response; a signed function can have the same first few moments.

The absence of an explicit delay in $m(b)$ can look surprising. The delay is present in the feasible set. A causal output kernel is supported in $[L, \infty)$, and an ordinary unit-mass function cannot place all its mass at the single point L. Thus

$$m(b) - L = \int_L^\infty (t-L) g_{b,c}(t)\ \mathrm{d}t > 0, \qquad b < 1 + \frac{1}{kK_p} - \frac{L}{T_i}.$$

This necessary restriction rules out some weights before any sign calculation, but it does not establish feasibility for the others.

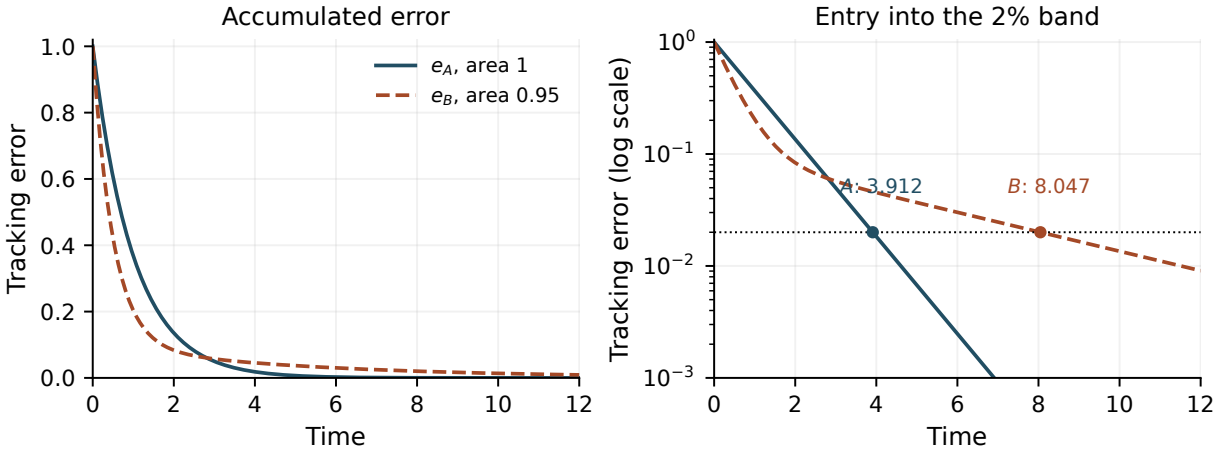


Figure S2: A smaller accumulated error can coexist with later settling. Both errors decrease to zero and generate monotone unit-gain responses. The second has area 0.95 instead of 1, but reaches the two-per-cent band at approximately 8.047, compared with 3.912 for the first. The logarithmic panel makes its slower tail visible.

6.1 A mean is not a threshold time

For $0 < \varepsilon < 1$, a monotone response has settled by time H exactly when $y(H) \geq 1 - \varepsilon$. With $g_{b,c} = g_0 + bg_b + cg_c$, this becomes

$$\int_0^H g_0(t)\,\mathrm{d}t + b\int_0^H g_b(t)\,\mathrm{d}t + c\int_0^H g_c(t)\,\mathrm{d}t \geq 1 - \varepsilon. \tag{6.7}$$

For fixed H it is one affine inequality added to $\mathcal{F}$. Feasibility persists when H increases. This is why Paper Proposition 16 uses bisection in time with a convex feasibility problem at each trial. It is a different calculation from maximising b: a response with a smaller mean can still have a longer small tail. For a concrete example, consider the two nonnegative errors

$$e_A(t) = e^{-t}, \qquad e_B(t) = \frac{9}{10}e^{-2t} + \frac{1}{10}e^{-t/5}.$$

Their negative derivatives are nonnegative unit-mass impulse kernels. The error areas are 1 and 19/20, respectively, so the second response has the smaller mean. But the first reaches its two-per-cent band at $H = \log 50$, while $e_B(H) \geq 50^{-1/5}/10 > 1/50$. The smaller-area response is still outside that band. This example illustrates a distinction within the class of positive unit-gain responses; it does not assert that both kernels belong to the same fixed PIDF family. Figure S2 displays the two error curves. The all-time constraints defining $\mathcal{F}$ remain part of every trial; checking only (6.7) would not establish permanent entry into the band.

The smallest feasible horizon is attained when $\mathcal{F}$ is nonempty. Choose feasible horizons H_n decreasing to their infimum and associated weights w_n. One feasible response eventually reaches the band, so that infimum is finite. Compactness gives a subsequence $w_n \to w_*$ in $\mathcal{F}$. The formula $y_w(H) = \int_0^H g_0 + b\int_0^H g_b + c\int_0^H g_c$ is jointly continuous in w, H. Passing to the limit retains the threshold inequality at the infimum. This is the existence step behind the bisection formulation, even though a finite sampled test alone cannot decide the required all-time positivity constraints.

6.2 How the MIACE projection enters

For a unit step and zero initial states and prehistory, put $v = ku - 1$, $A_e = \int_0^\infty |1 - y|$, and $A_u = \int_0^\infty |v|$. Both signals are integrable in the fixed stable loop. The command is measured

relative to its new equilibrium; the area of the uncentred nonzero steady command would be infinite. All times here use one fixed scale. The signed-area balance developed in Companion Section 6.1 is

$$\int_0^\infty e = m(b), \qquad \int_0^\infty v = \Gamma - m(b). \tag{6.8}$$

For Paper Theorem 17 assume additionally $\Gamma > 0$ and $\alpha > 0$. If $N = \int e_- + \int v_-$, then $A_e + A_u = \Gamma + 2N$, and elementary completion of the square gives

$$\mathcal{M}_\alpha^2 - \frac{\alpha\Gamma^2}{1+\alpha} = \frac{4\alpha N(\Gamma + N) + (A_e - \alpha A_u)^2}{1+\alpha}, \quad \mathcal{M}_\alpha^2 = A_e^2 + \alpha A_u^2. \tag{6.9}$$

The two nonnegative terms record two different losses: negative signal areas and an unbalanced division of the total area.

Monotonicity supplies $e \geq 0$. It does not supply $v \geq 0$. Impose the latter separately and call the resulting set $\mathcal{F}_+$. The centred step command is an affine L^1 function of the weights: its common steady part has been subtracted, and stability makes each remaining basis response integrable. The closed-positive-cone argument from Section 5 therefore makes $\mathcal{F}_+$ a closed convex subset of the compact set $\mathcal{F}$. When it is nonempty, its projection onto the b axis is a compact interval $[b_-, b_+]$. On this set both negative areas vanish, so

$$\mathcal{M}_\alpha^2 = (1+\alpha)\left(m(b) - \frac{\alpha\Gamma}{1+\alpha}\right)^2 + \frac{\alpha\Gamma^2}{1+\alpha}. \tag{6.10}$$

The unconstrained minimum asks for one particular error area. Since $m(b)$ is affine, its implementation is simply

$$b_\alpha = 1 + \frac{1}{kK_p} - \frac{\alpha\Gamma}{(1+\alpha)T_i}, \qquad \widehat{b}_\alpha = \min\{b_+, \max\{b_-, b_\alpha\}\}. \tag{6.11}$$

Any c with $(\widehat{b}_\alpha, c) \in \mathcal{F}_+$ attains the minimum on $\mathcal{F}_+$. If b_α lies inside the interval, the universal bound in (6.9) is attained. If it lies outside, clipping gives the constrained minimum, which is strictly larger. Allowing commands of changing sign requires their additional absolute area; the clipped formula alone no longer solves that larger problem.

7 Following the exact delayed example

Paper Theorem 18 gives a case in which all these requirements can be verified on the whole half-line. It starts with

$$P_L(s) = \frac{e^{-s/4}}{(1+s)^2}, \qquad C_y(s) = \frac{(1+s)^2}{s(as+d)}, \qquad d = \frac{27}{16}, \quad E_0 = e^{1/4}, \quad a = d - E_0 > 0. \tag{7.1}$$

Set $\rho = a/d$. The fixed D-branch PIDF parameters are

$$T_f = \rho, \quad T_i = 2 - \rho, \quad K_p = \frac{2-\rho}{d}, \quad T_d = \frac{(1-\rho)^2}{2-\rho}. \tag{7.2}$$

Only the weights move:

$$b(q) = \frac{1+q-\rho}{2-\rho}, \qquad c(q) = \frac{q-\rho}{1-\rho}, \qquad \frac{13}{16} \leq q \leq 1. \tag{7.3}$$

Substitution in the reference numerator gives

$$C_r(s) = \frac{(1+s)(1+qs)}{s(as+d)}, \qquad \frac{C_r}{C_y} = q + \frac{1-q}{1+s}.$$

Thus the new reference response combines the $q = 1$ response with a lagged copy. This identity describes the same controller weights; it does not add a filter state to the implemented controller.

7.1 The inverse with a positive kernel

On $[0, 1/4]$, define $J(t) = E_0 e^{-t}$ and $K(t) = E_0 e^{-t} - 1$, and set them to zero elsewhere. Both are nonnegative. Direct integration, with hats denoting Laplace transforms, gives

$$R(s) = as + E_0 - \widehat{J}(s) = 1 + as + s\widehat{K}(s), \qquad s(as + d) + e^{-s/4} = (1 + s)R(s). \tag{7.4}$$

Let $\gamma = E_0/a$ and $h(t) = e^{-\gamma t}/a$. Then

$$r = \mathcal{L}^{-1}(R^{-1}) = h * \left(\delta_0 + \sum_{n=1}^{\infty} (h * J)^{*n}\right), \qquad \|h * J\|_1 = \frac{E_0 - 1}{E_0} < 1. \tag{7.5}$$

This is precisely the convolution geometric series of Companion Sections 3.3–3.5. Every term is nonnegative, while the strict norm bound makes the series converge. A sufficiently small positive exponential weight preserves that strict bound, because J has finite support and h decays exponentially. The weighted series excludes zeros of R in a strip containing the closed right half-plane. The retained characteristic factor is proportional to $(1 + s)^3 R(s)$. Its remaining factors are stable, so the retarded-system criterion stated in Companion Section 5.4 gives internal exponential stability. This step checks the retained states, including their initial histories.

The simplified transfers are

$$G_q = \frac{e^{-s/4}(1 + qs)}{(1 + s)^2 R}, \qquad U_q = \frac{1 + qs}{R}, \qquad V_q = \frac{U_q - 1}{s} = \frac{q - a - \widehat{K}}{R}. \tag{7.6}$$

Here U_q maps the reference to the command, whereas V_q is the Laplace transform of the centred *step* command. At $q = 1$ the output impulse is a delay followed by convolution of e^{-t} and r, hence is nonnegative. The factor $q + (1 - q)/(1 + s)$ has a nonnegative unit-mass impulse measure, so the same is true for all the displayed q. Since $R(0) = 1$, every output has unit steady gain.

7.2 The separate command-sign calculation

The output argument leaves one sign unresolved. Differentiating the renewal equation for r gives $r' = -\gamma r + (J * r)/a \geq -\gamma r$ almost everywhere for $t > 0$. Absolute continuity makes $e^{\gamma t} r(t)$ nondecreasing, and hence $r(t - u) \leq e^{\gamma u} r(t)$ for $0 \leq u \leq t$, and

$$v_q = (q - a)r - K * r \geq (q - a - \kappa)r, \qquad \kappa = \int_0^{1/4} e^{\gamma u} K(u)\, \mathrm{d}u. \tag{7.7}$$

Elementary exponential-series bounds give

$$\frac{5}{4} < E_0 < \frac{9}{7}, \qquad \frac{45}{112} < a < \frac{7}{16}, \qquad \frac{\gamma}{4} < \frac{4}{5}, \qquad \int K = E_0 - \frac{5}{4} < \frac{1}{28}.$$

For example, $n! \geq 2^{n-1}$ for $n \geq 2$ bounds the part of $e^{1/4}$ after its linear term by the geometric sum $\sum_{n=2}^{\infty} 2(1/8)^n = 1/28$; at least one term is strictly smaller. Subtracting from d gives the bound on a, and their ratio gives $\gamma/4 < 4/5$. These rational estimates leave a comfortable margin, so rounded numerical values are unnecessary in the sign proof. Using $e < 3$ gives $\kappa < 3/28$. For $q \geq 13/16$, $q - a > 3/8$, and therefore $q - a - \kappa > 15/56 > 0$. This proves the command sign for every time, including the tail.

Now evaluating $V_q(0)$ computes an absolute area, because that sign has been proved. With $\Gamma = 2 + 1/4 = 9/4$,

$$A_u = q - \frac{7}{16}, \qquad A_e = \frac{43}{16} - q, \qquad q_\alpha = \frac{43}{16} - \frac{9\alpha}{4(1 + \alpha)}. \tag{7.8}$$

For $3 \leq \alpha \leq 5$, the last expression belongs to $[13/16, 1]$ and makes $A_e = \alpha A_u$. Both terms on the right of (6.9) vanish. Thus the attained value $(9/4)\sqrt{\alpha/(1 + \alpha)}$ is global among the settling zero-initial-state, zero-prehistory plant responses with integrable error and centred command, rather than merely best on this weight path.

7.3 Why the endpoint also beats overshooting competitors in IAE

For an arbitrary $(b, c) \in [0, 1]^2$, the fixed stable loop still gives an integrable tracking error. The signed-area identity therefore yields

$$\int_0^\infty |e_{b,c}| \geq \int_0^\infty e_{b,c} = (2 - \rho)(1 - b) + \frac{27}{16} \geq \frac{27}{16}. \tag{7.9}$$

At $q = 1$, both weights equal one and the established monotone response attains 27/16. No monotonicity assumption was imposed on the competitors in (7.9). That is the step which extends IAE optimality to the entire box, including overshooting responses. It does not establish uniqueness of c there.

Along the path itself there is also a stronger time-domain ordering. If $z(t) = \int_0^t e^{-(t-r)} y_1(r)\, \mathrm{d}r$, monotonicity gives $z(t) \leq y_1(t)$, and $y_q = qy_1 + (1 - q)z$ increases with q at every time. Increasing q therefore accelerates every threshold along this path. At $q = 13/16$ the areas are $(15/8, 3/8)$; at $q = 1$ they are $(27/16, 9/16)$. The error area falls by ten per cent while the extra-command area rises. These endpoints minimise MIACE for $\alpha = 5$ and $\alpha = 3$, respectively. Their disturbance and noise channels are identical. The improvement in tracking is paid for in command excursion, while the fixed feedback remains the same.

8 A reading map for all eighteen results

The numbering below is that of the MCSS version of Paper 14 prepared on 23 October 2026. "C13" means *Mathematical Background for MIACE and Monotone Tracking*, version 1.1 of 22 October 2026 [2]. Its section numbers are independent of the section numbers in this guide. Table S1 identifies the step to understand before returning to the paper's proof. It is a reading aid; the full hypotheses remain those of the paper.

Table S1: Reading map for the eighteen numbered results of the article.

Paper result	Step in the proof	Reading route
1. Proposition. Separation of the fixed feedback channels	Set the reference to zero in the shared-state realisation. Then solve the disturbance and noise equations. Both calculations remove the weights; the reference transfers retain them.	Guide §2; C13 §5.3.
2. Theorem. Exact constrained assignment	Invert the two coefficient equations. The filtered-D map is invertible and affine; the whole-output-filter map is diagonal. Applying the weight bounds to these inverses proves both directions of feasibility.	Guide §2.
3. Theorem. Cross-placement command-jump identity	Compute the right-hand command value for an unsmoothed step, with zero initial states and a strictly proper plant. Substitution of the matching weights reduces both values to $K_p \pi \Delta r / (T_i T_f)$.	Guide §2; C13 §§3.3, 5.3.
4. Corollary. Exact target under a jump limit	Take the absolute value in Result 3 and divide by positive constants. The bound concerns the initial jump; later command peaks require their own estimates.	Guide §2; Paper Result 3.

Continued on the next page

Paper result	Step in the proof	Reading route
5. Corollary. A necessary damping and frequency bound	Substitute $\sigma = 2\zeta_z/\omega_z$ and $\pi = \omega_z^{-2}$. The jump limit supplies a lower frequency bound and $b \geq 0$ an upper one. Their compatibility gives the damping bound; $b = 0$ gives sharpness under the stated restrictions.	Guide §2; Paper Results 2, 4.
6. Theorem. Exact reference-channel cancellation	Coprimality makes disappearance of the selected roots equivalent to divisibility by their quadratic factor. The constant coefficient one fixes the scaling. The cancelled stable states remain in the implementation.	Guide §§2, 3; C13 §5.3.
7. Theorem. Tight disk mismatch bound	Write $p = \widehat{p} + e$ in the normalised quadratic and apply the triangle inequality. Aligning the displacement gives equality on the unrestricted disk, which may be larger than the actual uncertainty family.	Guide §3.
8. Corollary. Uniform modal-residue bound	At a simple root, divide the numerator by the derivative of the denominator. Multiply Result 7 by a bound on this prefactor. Uniformity requires the stated pole separation and prefactor control.	Guide §3; C13 §4.3.
9. Corollary. Step-mode mismatch bound	The step adds a factor $1/s$, so the modal amplitude is the transfer residue divided by the pole. Combine the conjugate pair, then use the lower bounds on pole modulus and decay rate.	Guide §3; Paper Result 8; C13 §§4.1, 6.4.
10. Proposition. Finite-scenario convexity and existence	Separate real and imaginary parts of each affine modal amplitude. Its norm bound is a second-order cone constraint. Continuity on the nonempty compact weight polygon gives an attaining minimiser.	Guide §4.
11. Proposition. A command limit throughout a time interval	Integrate a uniform derivative bound from the nearest endpoint. Every point lies within half an interval length of one endpoint. The resulting margin fills the gaps between sampled constraints.	Guide §3; C13 §2.5.
12. Proposition. A sufficient family-wide feedback condition	Apply the quartic Routh–Hurwitz inequalities. Expand the determinants so that the uncertainty-dependent terms have known signs; the stated parameter bound makes the remaining term strictly positive.	Guide §2.2; Paper §8.2; C13 §5.3.
13. Proposition. Exact preservation of the feedback channels	Repeat Result 1 with the actual delayed state and history system. Identical disturbance and noise operators have identical norms. The reference-to-output and reference-to-command operators still vary.	Guide §§2, 5; C13 §§5.3–5.4.
14. Theorem. Fastest mean response with a fixed delayed feedback loop	Positivity and unit mass give monotonicity and unit reference H_∞ norm. Tonelli identifies IAE with the first moment. The low-frequency expansion makes that moment affine in b; independence of the two kernel directions bounds the feasible weights.	Guide §§5–6; C13 §§2.4–2.5, 3.1, 4.1–4.2.
15. Proposition. Second moment and a secondary choice of c	Retain the next term of the low-frequency expansion. Tonelli turns it into time-weighted error. At fixed optimal b, strict increase in c selects the smallest feasible c; nonnegative variance gives a necessary moment bound.	Guide §6; C13 §§2.4, 4.2.

Continued on the next page

Paper result	Step in the proof	Reading route
16. Proposition. Exact settling-time feasibility	For a monotone response, reaching the threshold at time H means remaining in the band thereafter. The threshold condition is affine in the weights. Compactness and joint continuity give attainment of the smallest feasible horizon.	Guide §6; Paper Result 14.
17. Theorem. MIACE reduction within the fixed-feedback family	Establish the signed-area balance, then account for negative parts before replacing signed areas by absolute areas. On the nonnegative-command subset, complete a scalar square and project the desired b onto the attainable interval.	Guide §6; C13 §§2.1, 6.1–6.2.
18. Theorem. A continuum of exact MIACE optima using only b, c	Prove stability of the retained delayed realisation and positivity of the constructed signals. Their explicit areas attain the MIACE bound. At the endpoint, the signed-error lower bound proves minimum IAE over the entire admitted weight box, including nonmonotone competitors.	Guide §7; C13 §§3.4–3.5, 4.5, 5.3–5.4, 6.1.

The unnumbered arguments also matter. The complete modal tail in Paper §5.1, the supporting-plane bound and parameter cover in Paper §6, and the verification discussion in Paper §10 explain how far a numerical calculation can support the analytical result. Sections 3–4 of this guide give the corresponding estimates. The family-wide stability proof in Result 12 does not establish pole simplicity, pole separation, or a continuum residue maximum; each needs the additional assumptions used in its own argument.

The separate manuscript-to-Lean map identifies formal declarations and records differences between the written and formal proof routes. This reading map instead identifies mathematical prerequisites. The additional explanations and examples in the present guide are not a new Lean development, and its creation does not enlarge the scope of the existing formalisation. A reader checking a formal theorem should compare its full hypotheses with the corresponding statement in the paper.

References and further reading

[1] L. P. Tomov. *Constrained Setpoint-Zero Shaping in Two-Degree-of-Freedom PIDF Control: Modal Suppression, Monotone Tracking, and MIACE with Pure Delay.* Manuscript, MCSS preparation version, 23 October 2026. Results 1–18.

[2] L. P. Tomov. *Mathematical Background for MIACE and Monotone Tracking: A Companion to the Proofs.* Version 1.1, 22 October 2026. Chapters 1–8. https://github.com/luchesart/MIACE/blob/monotone-tracking/MIACE_Proofs_Companion.pdf.

[3] L. P. Tomov. *MIACE and Monotone Tracking: Exact Tuning for First-Order Plants and Positive Cascades with Pure Delay.* Manuscript submitted to the IMA Journal of Mathematical Control and Information, 2026. Supporting formal proofs and map: https://github.com/luchesart/MIACE/tree/monotone-tracking.

[4] S. Boyd and L. Vandenberghe. *Convex Optimization.* Cambridge University Press, 2004. Chapters 2–5 provide a broader treatment of convex sets, functions, optimisation and duality. The authors make the book freely available at https://web.stanford.edu/~boyd/cvxbook/.

Preparation note. ChatGPT assisted with the exposition, calculations and LaTeX preparation. Responsibility for the text remains with the author. The additional illustrative calculations in this guide are not presented as newly checked Lean declarations.